SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I Navin kumar. M (192511015) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navin kumar. M

22+22+17+17+8

86

1. Smart Medical Diagonosis system

a) propositional logic Representation

Let:-

- F = patient has Fever
- C - patient has Cough
- B - patient has Breathlessness
- M - patient has Measles
- P - patient has pneumonia

Rules:

1. Fever AND Cough - possible Measles $F \wedge C \rightarrow M$

2. Fever AND Rash - possible Measles $F \wedge R \rightarrow M$

3. Cough AND Breathlessness → possible pneumonia

   $C \wedge B \rightarrow F$

Facts for patient A:

- F = True
- C = True
- B = True

b) Forward chaining

Fever (A), Cough (A), Breathlessness (A)

step 1:-

Fever (A) $\wedge$ Cough (A)

$\rightarrow$ Measles (A)

step 2:-

Cough (A) $\wedge$ Breathlessness (A)

$\rightarrow$ pneumonia (A)

Therefore, possible diagnoses are:

Measles and pneumonia

c) Backward chaining

Goal: pneumonia (A)

Rule:-

Cough (A) $\wedge$ Breathlessness (A) $\rightarrow$ pneumonia (A)

Goal reduction

Pneumonia (A)

↑

cough (A) AND Breathlessness (A)

↙ ↘

cough (A) Breathlessness (A)

✓

Both required facts are available

Therefore, pneumonia (A) is proved

2. Autonomous Traffic Management

a) Unification

Rule:

Vechicle (x) ∧ Emergency Type(x) → clearpath(x)

For Ambulance

MGU = {x / Ambulance}

Therefore: clearpath (Ambulance)

b) Resolution

- Vechicle (Ambulance) + Emergency Type (Ambulance) → clearpath (Ambulance)

- clearpath (Ambulance) → Green signal (Ambulance)

- Vechicle (car A) + Behind (Car A, Ambulance) + Green signal (Ambulance) → Proceed (Car A)

Hence,

Proceed (Car A) is proved

Current KB can represented multiple vechicles, but it lacks collision avoidance and priority rules.

3. Agricultural Expert system

a) CNF

1. $\neg$ soilDry $\wedge$ IrrigationNeeded

2. $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod.

3. ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

4. ~~soilDry~~

5. CropWheat

b) Resolution

soilDry $\rightarrow$ IrrigationNeeded

IrrigationNeeded + CropWheat $\rightarrow$ ApplyDrip Method.

Therefore

Apply Drip Method is proved

c)

If soil dry is removed

Irrigation Needed cannot be derived.

so Apply Drip Method cannot be concluded

Crop At Risk also cannot be definetely

concluded.

4. Wumpus World.

a) Modus ponens

- stench [1,2] - Wumpus adjacent to [1,2]
- Breeze [1,1] - pit adjacent to [1,1]
- Glitter [2,2] - Gold at [2,2]

b) possible location

- Wumpus [1,1], [1,3], [2,2]
- pit [1,2], [2,1]

c)

Path

[1,1] → [1,2] → [2,2]

is not guaranted safe. because [1,2] may contain a pit and [2,2] may contain the wompus.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

22 + 22 + 17 + 17 + 8

= 86

100